

NMAM INSTITUTE OF TECHNOLOGY, NITTE
(An Autonomous Institution affiliated to VTU, Belagavi)
Sixth Semester B.E. (CCE) (Credit System) Degree Examinations

May - June 2023

20CC601 – CYBER SECURITY AND FORENSICS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.*

		Marks	BT*	CO*	PO*
Unit – I					
1.	a) Explain the components of CIA triad with respect to the security of the system.	6	L*2	1	1
	b) Explain how the attacker attacks the substitution cipher with the understanding of frequency of letters in English language.	6	L2	1	1
	c) With the help of a neat diagram, explain the basic architecture of a firewalled network.	8	L2	1	1
2.	a) How can we improve the stream ciphers with block ciphers? Explain.	6	L2	1	1
	b) What are the components of a DNS? Explain.	6	L2	1	1
	c) With the help of an example, explain the process involved in RSA algorithm.	8	L3	1	2
Unit – II					
3.	a) With the proper message format, explain HTTP request and reply message.	10	L2	2	1
	b) What are the various fraud techniques? Explain in detail.	10	L2	2	1
4.	a) Why do the attackers use proxies? Explain how we can detect the use of proxies.	6	L2	2	1
	b) How to communication is carried in domain protocol? Explain with proper message format.	6	L2	2	1
	c) What is rogue antivirus? How will it effect on the various classes of users? Explain.	8	L2	2	1
Unit – III					
5.	a) Differentiate between a virus and a worm with suitable examples.	6	L2	3	1
	b) Explain any three persistent software techniques.	6	L2	3	1
	c) What are rootkits? Explain the categories of rootkits.	8	L3	3	1
6.	a) Explain virtual machine obfuscation.	10	L2	3	1
	b) How to detect the auto start entries? Explain.	10	L2	3	1
Unit – IV					
7.	a) What do you mean by form grabbing? Explain with suitable example.	10	L2	4	1
	b) How do you detect and prevent man-in-the-middle attack? Explain.	10	L2	4	1
8.	a) What is DLL injection attack? Explain the same with suitable illustration.	10	L3	4	1
	b) What you have understood about computer forensics and digital forensics? Elaborate the same.	10	L2	4	1
Unit – V					
9.	a) Explain why memory forensics are important with respect to computer's physical memory.	10	L2	5	1
	b) Explain why do we need cyber laws.	10	L2	5	1
10.	a) What are the weak areas of IT Act? Elaborate the same.	10	L2	5	1
	b) What are honeypots? Explain with the help of a neat figure, the honeynet structure.	10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
